Entrydesk: approach and next steps

I built a TypeScript workspace that turns shipment PDFs into an auditable NetCHB entry draft. The pipeline separates extraction, reconciliation, review and XML serialization. React presents shipment facts, a valuation bridge, source text and a completion checklist; Express validates uploads and holds short-lived review sessions in memory. This separation makes a live change to a field mapping or validation rule straightforward.

Contract discovery. I located NetCHB's public entry XSD, shared data types, example XML and EntryUploadService WSDL. The SOAP operation takes credentials and an XML string; the entry payload uses the vendor namespace and ordered root elements. I retained source snapshots and validate generated payloads offline against the real XSD. Missing required header fields stay missing, so an incomplete draft fails validation honestly. Six-digit invoice HS codes can pass the schema's length pattern but still fail the app's ten-digit US HTS completion check. No transmission or precalculated-duty flag is emitted.

Extraction without an API key. PDF.js reads digital text and renders scanned pages for local Tesseract OCR. Deskewing, thresholding, ruled-table detection and cell-level OCR recover all five sample packing rows. Extracted data and page-marked text retain document provenance. The parser uses labels and table headers rather than shipment-specific identifiers. Document-backed operational fields show their sources; conflicts, missing details and verification checks are separate, with related pricing evidence grouped. Unknown layouts can be corrected in a schema-validated JSON editor. An optional structured OpenAI vision adapter is implemented, but not live-tested because no API key was available.

Judgment on this shipment. The invoice's first line says 2,400 x $2.85 but prints $6,480 rather than $6,840. Commercial line amounts sum to $22,875, $360 below the printed $23,235 subtotal. Hoodie quantity and fiber composition disagree with the packing list; gross weight is 1,930 kg on the BOL versus 1,888 kg on the packing list. The blouse has a $4,180 buyer-supplied fabric assist, while $24 of free samples have Bangladesh origin. The line-based proposed value is $27,079, pending corrections and valuation confirmation. I preserve disagreements and require resolution instead of guessing. Factory MIDs, tax IDs, operational entry details, full classifications, statistical quantities, current remedies/PGA requirements and confirmed arrival remain broker tasks.

Validation and limits. Regression tests exercise extraction, disputed values, sample/assist handling, origin preservation and actual XSD validation. An HTTP integration test generates a different PDF shipment and completes it through upload, review and valid XML export; malformed PDFs and negative amounts are rejected. A second integration fixture verifies an unseen scanned table. Handwriting and some description text still need verification. Native browser permissions prevented visual UI testing. No live NetCHB upload, tariff calculation, arbitrary-layout accuracy claim or deployed service is asserted.

With more time. I would expand table recovery to borderless layouts, add source bounding boxes and measure accuracy across unseen documents, then support multi-invoice shipments. Next would be broker-confirmed HTS enrichment, date-specific tariffs/PGA rules, exchange rates and assist allocation. Production work would add isolated extraction workers, authenticated persistent review/audit storage and a NetCHB sandbox round trip. Finally I would complete browser/accessibility tests and deploy an authenticated demo.